Comparison of prostate specific antigen test outcomes for prostate cancer diagnosis in different ethnic groups – the EPIC study

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Background

- The Prostate Specific Antigen (PSA) blood test is a diagnostic tool for men with lower urinary tract symptoms
- Average PSA levels vary for men of different ethnic groups
- Prostate cancer incidence and mortality in the UK are highest for Black men and lowest for Asian men

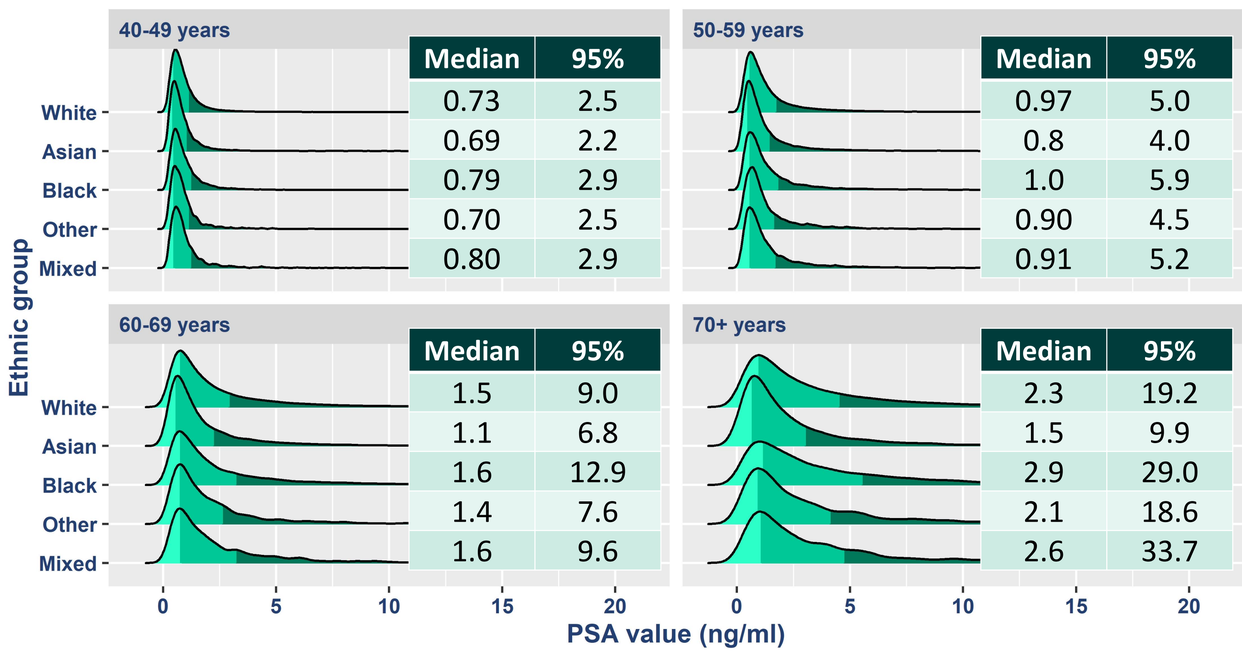
This study investigated differences in the diagnostic performance of PSA by patient ethnicity

Table 1. Cohort characteristics

Ethnic group	Number of patients	Percentage	Age-standardised prostate cancer rate/1000
White	649 445	88.9%	29.65
Asian	37 827	5.2%	12.15
Black	31 053	4.3%	45.23
Other	6 454	0.9%	22.03
Mixed	5 736	0.8%	32.05

Figure 1. PSA distribution by ethnic group and age category

Colour changes indicate 25% and 75% values, numerics for median and 95% values



Methods

- Anonymised electronic health record data from English general practice, cancer registrations, and secondary care

Included patients:

- Had a PSA test between 2010 and 2017
- Were aged at least 40 years old

Threshold for raised PSA:

- 40-49 year olds 2.5ng/ml
- 50-59 year olds 3.0ng/ml
- 60-69 year olds 4.0ng/ml
- 70 years and older 5.0ng/ml

Analysis:

- Multi-level logistic regression

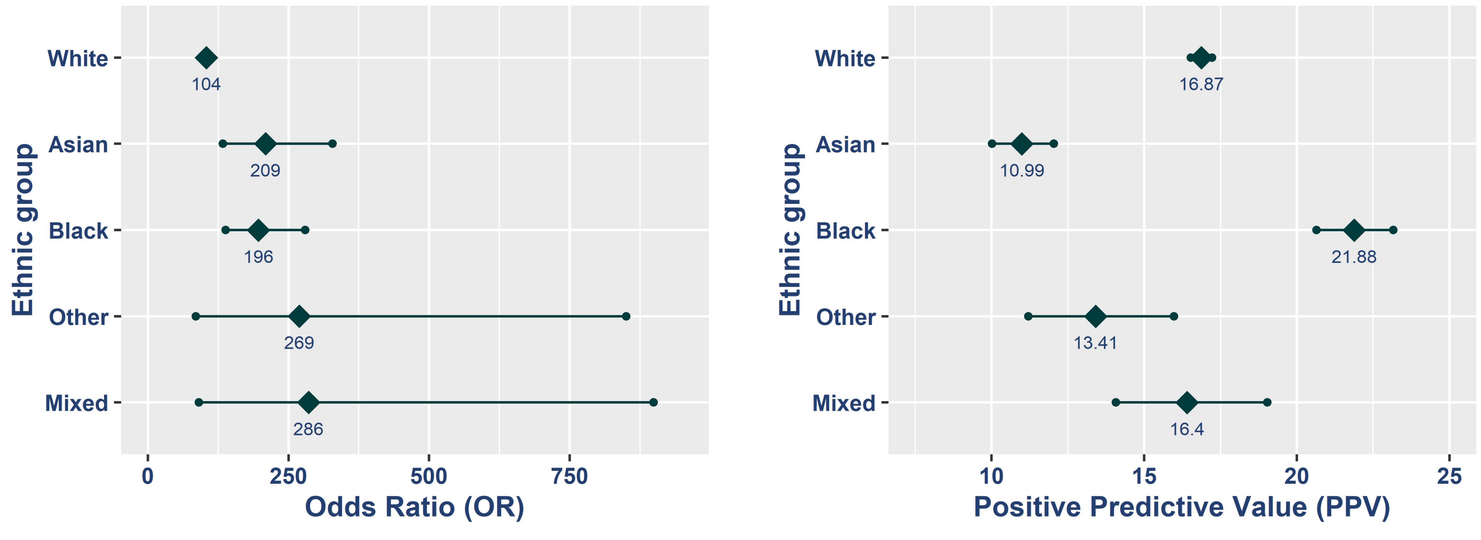
Results

- Black men had the highest prostate cancer incidence: 45 cases/1000 men
- Asian men had the lowest prostate cancer incidence: 12 cases/1000 men
- The odds of a prostate cancer diagnosis in Black men with raised PSA was higher than in White men
- In Black men, raised PSA had a PPV of 21.9
- In White men, raised PSA had a PPV of 16.9
- In Asian men, raised PSA had a PPV of 11.0
- Results for the Mixed and Other ethnic groups were unclear, largely because there were fewer men in these categories

Figure 2. PSA test performance in men from different ethnic groups

Adjusted for general practice, age, deprivation, and year of test

Showing 95% confidence intervals



Odds ratio for prostate cancer in men with a raised PSA result vs. men with a normal result

Positive Predictive Value of a raised PSA test for prostate cancer diagnosis

Conclusions

- The predictive utility of PSA tests for prostate cancer differ according to ethnic group, being highest in Black men, and lowest for Asian men
- The strength of association of a raised test result to cancer diagnosis also varies by ethnicity, being around twice as strong for Asian and Black men relative to White men
- Further research is needed to better understand the causes of these differences and the clinical implications

Acknowledgements

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Further information

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